Mathematics Bootcamp

Properties of Preferences and Utility Functions

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Today

- Indifference as an Equivalence Relation
- Utility Representation
- Adding Structure to Preferences
    - Monotonicity
    - Local Nonsatiation
    - Convexity
    - Continuity

Preference Relations

- A **preference relation** $\succeq$ is a binary relation on X: a subset of $X \times X$ recording the comparisons made by the DM. It need not compare every pair unless completeness is imposed.
- $x \succeq y$ reads: x is *at least as good* as y (x is *weakly better* than y).
- From $\succeq$ we derive:
  - Strict preference: $x \succ y \iff (x \succeq y$ and not $y \succeq x)$.
  - Indifference: $x \sim y \iff (x \succeq y$ and $y \succeq x)$.

Properties of Indifference

Indifference Relation

If $\succeq$ is rational (i.e., complete and transitive), then the induced indifference relation $\sim$ is:

1. **Reflexive**: $x \sim x$ for all $x \in X$.
2. **Symmetric**: $x \sim y \implies y \sim x$.
3. **Transitive**: $x \sim y \wedge y \sim z \implies x \sim z$.

Equivalence Relation

An equivalence relation is a binary relation that is reflexive, symmetric and transitive.

Equivalence Classes

Equivalence Classes

An equivalence relation induces a **partition** of the set: mutually disjoint subsets whose union equals the entire set.

If $\succeq$ is rational, its induced indifference relation $\sim$

1. is an equivalence relation;
2. generates equivalence classes of indifferent elements.

For each $x \in X$, the equivalence class of x is denoted by

$$[x] = \{y \in X \mid x \sim y\}.$$

Equivalence Classes and Utility

Utility representation

A function $u : X \to \mathbb{R}$ *represents* $\succeq$ if

$$x \succeq y \quad \Leftrightarrow \quad u(x) \geq u(y).$$

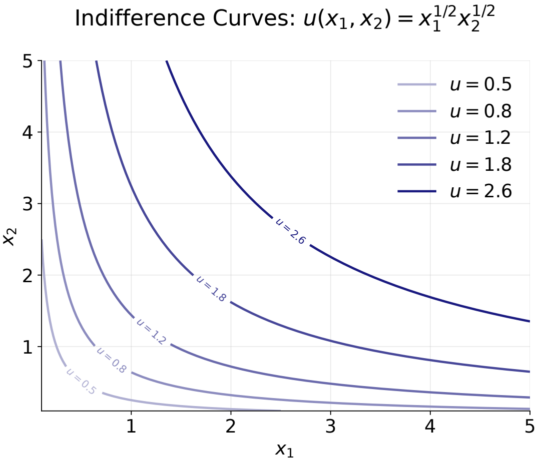
What the Representation Implies

If u represents $\succeq$, then

$$x \sim y \quad \Leftrightarrow \quad u(x) = u(y).$$

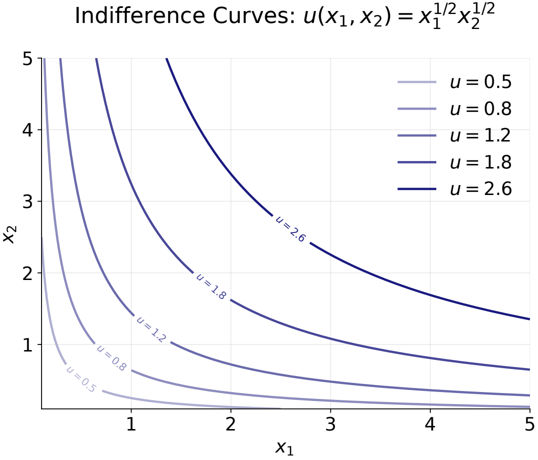
Thus, every equivalence class is mapped to a single utility level. Moreover, if $\phi : \mathbb{R} \to \mathbb{R}$ is strictly increasing, then $\phi \circ u$ represents the same preferences: utility is **ordinal**.

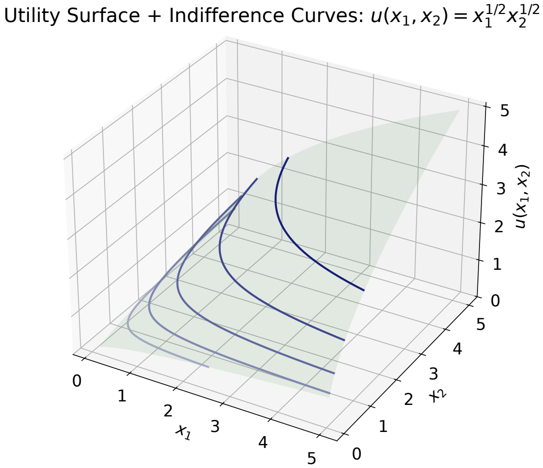
Indifference Curves



- Indifference curves are equivalence classes of X that map to the same utility in $\mathbb{R}$.
- Equivalence classes are disjoint, so indifference curves can never cross.

Utility Representation





Useful Properties of Preferences

- In addition to rationality, we need more structure in preference relations. That is, we need to assume more properties.
- These properties have implications for:
  - Their utility function representation.
  - Their indifference curves.
- The best way to understand and learn them is by practicing proofs, many times.
  - You will have plenty of Weekly Problem Sets where you will do exactly that.

Represent Commodities with Ordered Lists

2.B Commodities

The decision problem faced by the consumer in a market economy is to choose consumption levels of the various goods and services that are available for purchase in the market. We call these goods and services *commodities*. For simplicity, we assume that the number of commodities is finite and equal to L (indexed by $\ell = 1, \dots, L$).

As a general matter, a *commodity vector* (or commodity *bundle*) is a list of amounts of the different commodities,

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_L \end{bmatrix},$$

and can be viewed as a point in $\mathbb{R}^L$, the commodity *space*.[1]

We can use commodity vectors to represent an individual's consumption levels. The ℓth entry of the commodity vector stands for the amount of commodity ℓ consumed. We then refer to the vector as a *consumption vector* or *consumption bundle*.

- Coordinatewise order: $x \geq y$ iff $x_\ell \geq y_\ell$ for every $\ell \in \{1, \dots, L\}$.
- Hence, $x \geq y$ and $x \neq y$ imply $x_\ell > y_\ell$ for some ℓ.

Monotonicity: More is Better

Definitions

Let $X \subseteq \mathbb{R}^L$. A preference relation $\succeq$ on X is

weakly monotone if $x \geq y$ implies $x \succeq y$;

strictly monotone if $x \geq y$ and $x \neq y$ imply $x \succ y$,

for all $x, y \in X$.

A function $f : X \to \mathbb{R}$ is

non-decreasing if $x \geq y$ implies $f(x) \geq f(y)$;

strictly increasing if $x \geq y$ and $x \neq y$ imply $f(x) > f(y)$, for all $x, y \in X$.

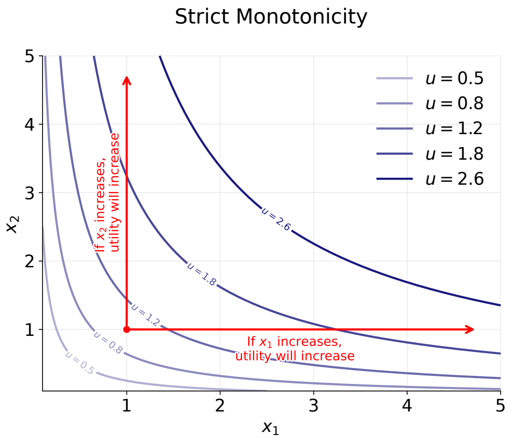
Proposition

If $u : X \to \mathbb{R}$ represents $\succeq$, then

$\succeq$ is weakly monotone *iff* u is non-decreasing;

$\succeq$ is strictly monotone *iff* u is strictly increasing.

Monotonicity: More is Better



Domain: $X = \mathbb{R}^2_{++}$. The function is not strictly increasing on the axes of $\mathbb{R}^2_+$.

Local Nonsatiation: No Thick Indifference Curves

Definition: Local Nonsatiation

A preference relation $\succeq$ on X is *locally nonsatiated* if for every $x \in X$ and every $\varepsilon > 0$, there exists $y \in X$ such that $\|x - y\| \leq \varepsilon$ and $y \succ x$.

Thus, preferences are locally nonsatiated if for each x you can find an arbitrarily close y that is strictly better.

Recall that we are in a Euclidean space, so

$$\|x - y\| = \sqrt{\sum_{i=1}^{L} (x_i - y_i)^2}.$$

Local nonsatiation rules out “thick” indifference sets.

Local Nonsatiation: No Thick Indifference Curves

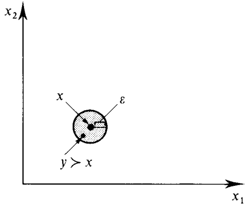


Figure 3.B.1
The test for local nonsatiation.

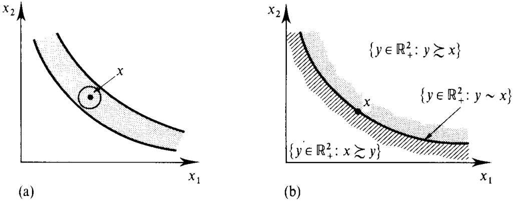


Figure 3.B.2
(a) A thick indifference set violates local nonsatiation.
(b) Preferences compatible with local nonsatiation.

Monotonicity and Local Nonsatiation

Proposition

On $X = \mathbb{R}^L_+$, strict monotonicity implies local nonsatiation.

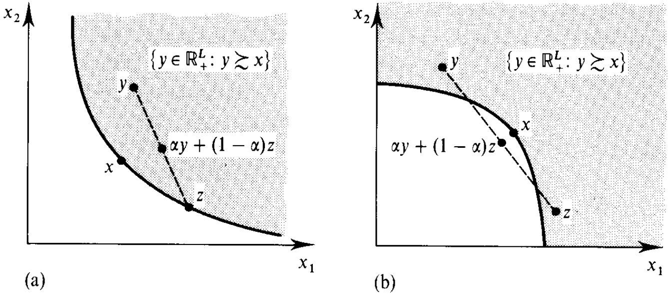
Proof. Suppose $\succeq$ is strictly monotone, i.e. $x \geq y$ and $x \neq y$ imply $x \succ y$. For each $x \in \mathbb{R}^L_+$ and every $\varepsilon > 0$, let $y = x + \frac{\varepsilon}{L}\mathbf{1}$. Then $y \in \mathbb{R}^L_+$ and

$$\|x - y\| = \sqrt{\sum_{i=1}^{L} \left(\frac{\varepsilon}{L}\right)^2} = \sqrt{L\left(\frac{\varepsilon}{L}\right)^2} = \frac{\varepsilon}{\sqrt{L}} \leq \varepsilon$$

Strict monotonicity implies that $y \succ x$.

However, local nonsatiation does not imply monotonicity.

Convexity: Taste for Variety



Definition 3.B.4: The preference relation $\succsim$ on X is *convex* if for every $x \in X$, the upper contour set $\{y \in X: y \succsim x\}$ is convex; that is, if $y \succsim x$ and $z \succsim x$, then $\alpha y + (1-\alpha)z \succsim x$ for any $\alpha \in [0, 1]$.

Figure 3.B.3(a) depicts a convex upper contour set; Figure 3.B.3(b) shows an upper contour set that is not convex.

Convexity: Taste for Variety

Convex Sets and Preferences

A set $X \subseteq \mathbb{R}^L$ is *convex* if $x, y \in X$ and $\alpha \in [0,1]$ imply $\alpha x + (1-\alpha)y \in X$.
Let X be convex. For each $x \in X$, define $U(x) := \{y \in X : y \succeq x\}$. A preference relation $\succeq$ on X is *convex* if

$$y, z \in U(x),\ \alpha \in [0,1] \quad \Longrightarrow \quad \alpha y + (1-\alpha)z \in U(x).$$

It is *strictly convex* if $y \neq z$ and $\alpha \in (0,1)$ imply $\alpha y + (1-\alpha)z \succ x$ whenever $y, z \in U(x)$.

Precise Geometric Statement

Convex preferences mean that every upper contour set $U(x)$ is convex. This is the precise statement behind the informal “taste for variety” description.

Level Sets and Quasiconcavity

Definitions

Let $X \subseteq \mathbb{R}^L$ be convex. A function $f : X \to \mathbb{R}$ is *quasiconcave* if, for $x, y \in X$ and $\alpha \in [0, 1]$,

$$f(\alpha x + (1 - \alpha)y) \geq \min\{f(x), f(y)\}.$$

It is *strictly quasiconcave* if, for $x \neq y$ and $\alpha \in (0, 1)$,

$$f(\alpha x + (1 - \alpha)y) > \min\{f(x), f(y)\}.$$

A More Transparent Definition

A function $f : X \to \mathbb{R}$ is *quasiconcave* if every upper level set of f is convex. That is, $\{\, x \in X \mid f(x) \geq c \,\}$ is convex for every $c \in \mathbb{R}$.

Convex Preferences and Quasiconcave Utility

Exact Equivalence

Let X be convex and let $u : X \to \mathbb{R}$ represent $\succeq$. Then

$$\begin{aligned} \succeq \text{ is convex} &\iff u \text{ is quasiconcave}, \\ \succeq \text{ is strictly convex} &\iff u \text{ is strictly quasiconcave}. \end{aligned}$$

Why

$$U(x) = \{y \in X : y \succeq x\} = \{y \in X : u(y) \geq u(x)\}.$$

Thus upper contour sets are upper level sets of u; replacing weak with strict inequalities gives the second equivalence.

Topology on the Choice Set

Relative Topology on $X \subseteq \mathbb{R}^L$

For $x \in X$ and $\varepsilon > 0$, the relative open ball is

$$B_\varepsilon^X(x) = \{y \in X : \|y - x\| < \varepsilon\}.$$

- A set $G \subseteq X$ is *open relative to* X if every $x \in G$ has some $\varepsilon > 0$ with $B_\varepsilon^X(x) \subseteq G$.
- A set $F \subseteq X$ is *closed relative to* X if $X \setminus F$ is open relative to X.

Why “Relative” Matters

At a boundary point of X—for example, where one commodity is zero—the ball is intersected with the feasible choice set X.

Lower and Upper Contour Sets

Contour Sets at x

$$U(x) := \{y \in X \mid y \succeq x\} \qquad L(x) := \{y \in X \mid x \succeq y\}.$$

The indifference set through x is always

$$I(x) := \{y \in X : y \sim x\} = U(x) \cap L(x).$$

Weak and Strict Sections

If $\succeq$ is complete, then

$$\{y \in X : y \succ x\} = X \setminus L(x), \qquad \{y \in X : x \succ y\} = X \setminus U(x).$$

Continuity: Equivalent Tests

Continuity via Contour Sets

Let $X \subseteq \mathbb{R}^L$ and let $\succeq$ be complete. We call $\succeq$ *continuous* if, for every $x \in X$, the weak contour sets $U(x)$ and $L(x)$ are closed relative to X.

Equivalent Open-Set Test

Under completeness, continuity is equivalent to requiring $\{y \in X : y \succ x\}$ and $\{y \in X : x \succ y\}$ to be open relative to X for every $x \in X$.

Interpretation

Small changes in a bundle cannot cause a strict ranking to reverse abruptly.

Continuity: Rankings Survive Limits

Equivalent Sequence Test

Let $X \subseteq \mathbb{R}^L$ and let $\succeq$ be complete and transitive. Continuity is equivalent to

$$(x_n, y_n) \to (x, y), \quad x_n \succeq y_n \text{ for every } n \quad \Longrightarrow \quad x \succeq y.$$

Equivalent Local Test

Equivalently, strict preference is open in $X \times X$: if $x \succ y$, there exist $\varepsilon_x, \varepsilon_y > 0$ such that

$$x' \in B^X_{\varepsilon_x}(x), \quad y' \in B^X_{\varepsilon_y}(y) \quad \Longrightarrow \quad x' \succ y'.$$

Debreu's Representation Theorem

Continuous Representation

Let $X = \mathbb{R}^L_+$. If $\succeq$ is complete, transitive, and continuous on X, then there exists a *continuous* utility function $u : X \to \mathbb{R}$ such that

$$x \succeq y \iff u(x) \geq u(y).$$

- A continuous utility representation implies continuous preferences.
- The theorem guarantees at least one continuous representation; it does not make every utility representation continuous.
- Rationality is necessary. On $\mathbb{R}^L_+$, continuity supplies enough additional structure for a continuous representation.